

Infection Control and Prevention of Disease Transmission in Oral Health Care

Kandis V. Garland

Preventing disease transmission in dental hygiene practice requires an understanding of microbial transfer, infection, and disease in the oral health care environment. Implementing and practicing the elements of standard precautions is essential.

Disease Transmission

See the section on “Microbial Virulence and Disease Transfer” in Chapter 9.

A. Development of an infectious disease

1. Source of microbe or pathogen (capable of causing infectious disease)
 - a. Primarily originating from a patient’s mouth
 - b. May originate from dental health care personnel (DHCP) (this mode less likely than from a patient’s mouth)
 - c. Microbes may be present in saliva, blood, respiratory secretions, and other bodily fluids
2. Portal of exit—escape of microbe from a person
 - a. Coughing, sneezing, and talking
 - b. Contaminated dental equipment, instruments, and supplies used in a patient’s mouth
 - c. Spatter droplets and aerosol particles generated by using power instrumentation, low-speed and high-speed handpieces, and air/water syringe
 - (1) Spatter droplets—large, visible particles that settle quickly, contaminating the clinician and the surfaces of the treatment area
 - (2) Aerosol particles—small, invisible particles that remain airborne for an extended period and may be inhaled
3. Transfer of microbes to another person
 - a. Direct contact—person to person exposure of microbes
 - b. Indirect contact—exposure to items contaminated by a patient’s microbes, including sharps (all items that can puncture skin), instruments, surfaces, hands, and equipment
 - c. Droplet infection—contact with spatter, spray, splashes containing microbes
 - d. Airborne infection—contact with aerosol particles containing microbes
4. Portal of entry—entry of microbes into a person
 - a. Inhalation—breathing aerosol particles
 - b. Ingestion—swallowing droplets of saliva or blood spattered into the mouth
 - c. Through mucous membranes—droplets of saliva or blood spattered into the eyes, nose, or mouth (ocular, nasal, oral)
 - d. Through breaks in the skin—touching patient tissues or contaminated objects, spattering of microbes onto nonintact skin, or punctures with contaminated sharps
 - (1) Percutaneous injury—injury that penetrates the skin
 - (2) Parenteral—intravenous or by injection
5. Infection
 - a. The infectious agent has a portal of entry into a susceptible (not immune) host
 - b. Establishment, multiplication, and survival of microbe in the body

6. Damage to the body—the disease occurs when microbes multiply to a harmful level
7. Pathogenic agents and diseases associated with oral health care (Table 10.1)

B. Infection control

1. Goal—to prevent oral health care–related infections, injuries, and illnesses among patients, DHCP, and community
2. Reducing the numbers of microorganisms that may be transmitted from person to person or from persons to the oral health care environment, and vice versa
3. Preventing disease transmission includes reducing circulating microbes while increasing immunity of the body
4. Infection control protocols seek to reduce or eliminate the likelihood of disease transmission by eliminating one or more factors required for disease transfer (Fig. 10.1)
 - a. Asepsis—absence of infectious materials; achieved by removing or killing microorganisms
 - b. Disinfection—reducing the number of pathogenic organisms in or on an object, thereby minimizing the potential for disease transmission
 - (1) Disinfectants are applied to inanimate objects
 - (2) Antiseptics are applied to living tissues
 - c. Engineering controls—devices that reduce the risk of exposure to potentially infectious materials (e.g., self-sheathing needles and instrument cassettes)
 - d. Personal protective equipment—items designed to protect DHCP from exposure to bloodborne and other pathogens (e.g., gloves, masks, N95 respirators, safety glasses, face shields, barrier gowns)
 - e. Standard precautions—treating all blood and bodily fluids (including secretions and excretions, except sweat), nonintact skin, and mucous membranes as potentially infectious in all patients
 - (1) In the oral health care setting, saliva is an important source of contamination; although invisible, it is capable of containing infectious agents that can survive on surfaces for a long period
 - f. Transmission-based precautions—additional measures beyond standard precautions provided during urgent care to patients with certain diseases having multiple modes of transmission (e.g., tuberculosis [TB]); in this situation, care would be provided in a facility equipped with a respiratory protection program including an airborne infection isolation room (AIIR)
 - g. Sterilization—the destruction or removal of all microorganisms in or on an object
 - h. Work practice controls—procedures that reduce the chance of exposure to potentially infectious materials (e.g., using a one-handed technique to recap needles; maintaining a 14-inch to 18-inch working distance to reduce exposure to droplets; fulcruming one to four teeth away from tooth being treated)
 - i. DHCP-related controls
 - (1) Maintaining a healthy lifestyle and immune status
 - (2) Receiving recommended vaccinations
 - (3) Obtain a baseline tuberculin skin test (TST)
 - (4) Abiding by the recommended US Public Health Service work restrictions for DHCP in the case of certain infections and after exposure to some diseases (Box 10.1). Consult with department of public health, CDC, and health care provider for return to work guidance
 - j. Patient-related controls

- (1) Prophylactic antibiotic premedication to reduce the possibility of infection in individuals with systemic health-related conditions, especially those with immune deficiencies
- (2) Preprocedural rinse to decrease the microbial load in spatter and aerosols and to help protect against autogenous (self-derived) infection
- (3) Avoiding treatment of patients with known communicable illnesses (e.g., influenza) or contagious lesions (e.g., herpes)
5. Infection control protocols are designed to reduce or prevent the spread of pathogens from:
 - a. Patient to DHCP
 - b. DHCP to patient
 - c. Patient to patient
 - d. Dental office to community, including families of DHCP
 - e. Community to patient
6. Table 10.2 outlines the mechanisms of the spread and the prevention of diseases
Table 10.1

Pathogenic Agents Important in Oral Health Care

Disease	Pathogen
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Bloodborne Diseases

Viral

Hepatitis B	Hepatitis B virus
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Hepatitis C	Hepatitis C virus
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Hepatitis D	Hepatitis D virus
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Acquired immunodeficiency syndrome	Human immunodeficiency virus
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Oral Diseases

Bacterial

Gonococcal pharyngitis	Neisseria gonorrhoeae
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Streptococcal pharyngitis and scarlet fever	Streptococcus pyogenes
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Syphilis	Treponema pallidum
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Viral

Primary herpetic gingivostomatitis	Human herpesvirus 1 or 2
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Recurrent herpes (e.g., herpes labialis)	Human herpesvirus 1 or 2
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Hand-foot-and-mouth disease	Coxsackievirus
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Herpangina	Coxsackievirus
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Hairy leukoplakia	Human herpesvirus 4
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Fungal

Candidiasis (thrush)	Candida albicans
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Denture stomatitis	C. albicans
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Systemic Diseases with Oral Lesions

Bacterial

Secondary syphilis	Treponema pallidum
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Viral

Chickenpox	Human herpesvirus 3 (Varicella-Zoster virus)
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Infectious mononucleosis	Human herpesvirus 4 (Epstein-Barr virus)
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Other Diseases Spread by Respiratory or Oral Fluids

Bacterial

Tuberculosis *Mycobacterium tuberculosis*

Diphtheria *Corynebacterium diphtheriae*

Pneumonia *Streptococcus pneumoniae*, *Staphylococcus aureus*, *Mycoplasma pneumoniae*, *Chlamydia pneumoniae*, *Moraxella catarrhalis*, *Haemophilus influenzae*

Meningitis, sinusitis, conjunctivitis *H. influenzae* type B

Meningitis *Neisseria meningitidis*

Bronchitis *H. influenzae*, *Moraxella catarrhalis*

Viral

Common cold Rhinoviruses and several others

Influenza Influenza viruses

Bronchitis Influenza A, parainfluenza virus, coronavirus

Pneumonia Influenza virus, adenovirus, respiratory syncytial virus

Cytomegalovirus disease Cytomegalovirus

Infectious mononucleosis Human herpesvirus 4 (Epstein-Barr virus)

Erythema infectiosum (fifth disease) Human parvovirus B19

Measles Rubeola (measles) virus

Rubella Rubella virus

Mumps Mumps virus

Middle eastern respiratory syndrome (MERS) Coronavirus (MERS-CoV)

COVID-19 SARS-CoV-1 (Coronavirus)

Severe acute respiratory syndrome SARS-CoV-2

Modified from Miller CH. Infection Control and Management of Hazardous Materials for the Dental Team. 7th ed. Elsevier; 2022.

FIG. 10.1 Factors related to infection and disease transfer. OHCW, Oral health care worker.

Follow for extended description

Box 10.1

Summary of Work Restrictions for Dental Health Care Personnel

Conjunctivitis

Diarrheal disease

Enteroviral infection

Hepatitis A

Hepatitis B and HIV (consult expert review panel)

Herpes on the hands

Measles

Meningococcal infection

Mumps

Pediculosis

Pertussis

Rubella

Viral respiratory infection (acute, febrile) (e.g., COVID-19)

Active Staphylococcus aureus infection
 Group A streptococcal infection
 Active tuberculosis
 Varicella (chickenpox)
 Zoster (shingles)
 From Guidelines for Infection Control in Dental Health-Care Facilities. 2003.
<https://www.cdc.gov/mmwr/preview/mmwrhtml/rr5217a1.htm#tab1>.

Table 10.2

Mechanisms of Disease Spread and Prevention

Pathway of Cross-Contamination	Source of Microorganism	Mode of Disease Spread
Mechanism or Site of Entry Into Body	Infection Control Procedure	
Patient to DHCP	Patient's mouth	Direct contact Through breaks in skin of dental team
Gloves and handwashing		

Immunizations

Droplet infection Inhalation by DHCP
 Facemask/N95 respirator

Rubber dam

Preprocedural rinse Face shield Avoid AGPs

Through breaks in skin of DHCP
 Gloves and handwashing

Protective clothing

Face shield

Rubber dam

Preprocedural rinse

Through mucosal surfaces of DHCP
 Facemask/N95 respirator

Eyewear

Face shield

Rubber dam

Preprocedural rinse

Immunizations

Airborne infection Inhalation by DHCP Facemask/N95 respirator/Face shield/PPR Rubber dam/Avoid AGPs

Indirect contact Cuts, punctures, or needlesticks in DHCP

Needle safety and waste management

Utility gloves for cleanup

Ultrasonic cleaning rather than hand scrubbing

Instrument cassettes to reduce direct handling during cleaning

Antimicrobial holding solution

Antimicrobial cleaning solution

Through breaks in skin of DHCP

Utility gloves for cleanup

Protective clothing

Immunizations

Patient's skin lesions Direct contact Through breaks in skin of DHCP

Gloves and handwashing

Immunizations

DHCP to patient DHCP hands (lesions or bleeding) Direct contact Through mucosal surfaces of patient

Gloves and handwashing

Care in handling sharp objects

Immunizations

Indirect contact Bleeding on items used in patient's mouth

Gloves and handwashing

Instrument sterilization

Surface disinfection

Immunizations

DHCP mouth (oral or respiratory fluids)
Facemask/N95 respirator

Droplet infection

Inhalation by patient

Face shield

Through oral mucosal surfaces of patient
Facemask/N95 respirator

Face shield

Patient to patient in light of COVID-19
inhalation
surfaces of patient
Instrument and handpiece sterilization

Patient's mouth
Indirect contact (instruments, surfaces, hands)

Patient's nose and mouth,
Through oral mucosal

High-volume evacuation systems

Air filtration systems

Avoid AGPs

Sterilization monitoring

Surface covers

Surface disinfection

Handwashing and proper gloving

Changing mask

Decontaminating protective eyewear

Changing protective clothing when needed

Use of sterile or clean supplies

Flushing dental unit water lines

Monitoring water line antiretraction valves

Use of disposable items

Use of one-way valves for slow-speed suction

OHC setting to community
PPR

Patient's mouth

Indirect contact

Anyone entering OHC setting may be exposed to COVID-19 including caregivers, interpreters, dental product/equipment representatives, repair and cleaning personnel, consultants and continuing education providers

Cuts, punctures, breaks in skin of dental laboratory worker, waste disposal, or laundry personnel

Waste management

Disinfection of impressions and appliances

Proper management of contaminated laundry

Handwashing

DHCP to DHCP's family DHCP bodily fluids Direct or indirect contact Intimate
contactImmunization

Community to patient Municipal water Direct contact or airborne Patient's mouth or
lungs

Independent water reservoir

Periodically disinfecting inside of dental unit water lines

Quarantine/isolationUse of water containing an approved antimicrobial agent

Filtering the water

AGPs, Aerosol-generating procedures; DHCP, dental health care personnel; OHC, oral health care; PPR, preprocedural rinse.

Modified from data from U.S. Department of Labor. Occupational Safety and Health Administration. Controlling Occupational Exposure to Bloodborne Pathogens. OSHA; 2001:3127 (revised).

Infection Control Procedures for Dental Health Care Personnel

A. Health history

1. Obtain, review, and update patients' health histories at all visits; obtain vital signs including blood pressure, pulse, respiration, and temperature
2. Address specific questions related to present health status, physician care, hospitalizations, surgery, diseases, medications, allergies, current or chronic illness, review of major organ systems, and recent overseas travel
3. Be aware that obtaining health histories will not identify all infectious patients; patients may suppress information purposely or unknowingly (many persons with hepatitis B virus [HBV], hepatitis C virus [HCV], and human immunodeficiency virus [HIV] are asymptomatic)
4. Health history may identify conditions requiring:
 - a. Diagnostic evaluation and laboratory testing
 - b. Medical consultations
 - c. Prophylactic antibiotic premedication to reduce the incidence of autogenous infections, that is, conditions caused by introducing the microflora of patients into injured tissues (e.g., bacteremia, abscess)
5. The US Centers for Disease Control and Prevention (CDC) recommends conducting a TB risk assessment for an oral health care setting and then formulating a prevention program appropriate for its designated risk category; most dental settings will be low or very low risk.

B. Immunizations are recommended for DHCP

1. Essential component of an infection control program
2. Most effective method to prevent contracting a vaccine-preventable disease
3. Immunizations are recommended for the following diseases, unless evidence of past infection and immunity: hepatitis A and B, varicella, influenza, measles, mumps, rubella, tetanus, diphtheria, pertussis, meningococcal disease, and COVID-19 vaccine when available
4. The US Occupational Safety and Health Administration (OSHA) policies regarding hepatitis B vaccination
 - a. OSHA requires employers within 10 working days to make the hepatitis B vaccine available at no cost to employees who are or may be exposed by occupation to bloodborne pathogens
 - b. Employers must provide information about the vaccine and ensure that all medical records concerning the vaccination are kept confidential
 - c. Employees have the right to refuse vaccination but must read and sign an OSHA declination statement
 - d. Employers cannot request employees be serologically prescreened before vaccination
5. Hepatitis B vaccine
 - a. Three single-antigen vaccines available:
 - (1) Recombivax HB (Merck & Co., Inc.)
 - (2) Engerix-B (GlaxoSmithKline)
 - (3) Heplisav-B (Dynavax Technologies)
 - b. One combination-antigen (hepatitis A and B) vaccine available:

(1) Twinrix (GlaxoSmithKline)

c. Process

- (1) Serologic screening for antibodies to hepatitis B surface antigen (HBsAg) before vaccination is not recommended, unless infection is suspected
- (2) Pregnancy is not a contraindication for vaccination
- (3) A series of three injections in the deltoid muscle given at 0, 1, and 6 months
- (4) Seroconversion rates are 95% to 97% in healthy younger adults; lower rates (approximately 70%) in persons over 40 years old, smokers, overweight persons, and those receiving injections in the buttocks
- (5) Genetic factors may influence seroconversion rates
- (6) Testing for antibodies (anti-HBsAg) is recommended after vaccination to ensure protection against HBV; testing should be conducted 1 to 2 months after the final injection
- (7) With successful seroconversion, protective antibodies have been sustained for at least 20 years
- (8) Failure to seroconvert requires a second series of three injections
- (9) Continued failure to seroconvert could signal chronic HBV infection in a person who is unable to produce antibodies; nonresponders should be evaluated by a medical provider
- (10) The CDC currently does not recommend a booster injection
- (11) Minimal side effects of the vaccine include injection site soreness, headache, and fever
- (12) Individuals with allergies to yeast or iodine (vaccine preservatives) should consult their primary care provider before the vaccination

6. Recommended screenings for DHCP

a. Hepatitis B blood testing to determine anti-HBsAg titer

b. TB testing in health care workers

- (1) Initial baseline testing upon hire: two-step testing with Mantoux TST or TB blood test
 - (a) Blood test: may be given to DHCP born outside the United States that have been given a vaccine called Bacille Calmette-Guérin (BCG). The BCG vaccine may cause a false-positive TST. Blood tests are not affected by prior BCG vaccination and are not expected to give a false-positive result
- (2) Annual or serial screening: determined by state regulations or risk assessment outcomes (Table 10.3)

c. Majority of dental offices are at low risk for DHCP contracting TB

- (1) Oftentimes, a positive test indicates previous infection with *Mycobacterium tuberculosis* but not active disease

d. Anti-TB drug therapy advised to prevent future active disease

- (1) Protocols should be in place to identify patients with active disease so they may be referred for treatment in a hospital-based facility
- (2) TB is acquired by inhaling respiratory droplets from an infected person
- (3) Only 10% of individuals infected will eventually develop disease
- (4) TB has become problematic for certain populations, including the homeless, drug abusers, and persons infected with HIV
- (5) Multiple drug-resistant TB has emerged in hospitalized and institutionalized individuals and those with AIDS

C. Hand hygiene

1. Extremely important disease prevention practice
2. Hands are a primary source of microorganisms capable of disease transmission
3. Rationale
 - a. Handwashing reduces both resident and transient flora on skin
 - (1) Resident flora are permanent residents that colonize several skin layers and cannot be completely removed; this type is less important and less likely to transmit disease than transient flora
 - (2) Transient flora contaminate hands when hands touch contaminated surfaces; this type colonizes the outer layers of skin, only survive for a limited time, and can be easily removed by routine handwashing
 - b. Protects patients and DHCP by reducing the spread of microorganisms and subsequent infections by the hands
 - c. Unwashed, contaminated hands can transfer microbes to sterile instruments, dental equipment, and environmental surfaces
4. Use of gloves is not a substitute for routine handwashing
5. Hands must be washed before gloves are put on to minimize organisms that can multiply rapidly when enclosed in a moist, warm environment; bacteria and yeast growth can cause skin irritation
6. Hands must be washed after removal of gloves because defects, tears, and punctures may occur in gloves, permitting microbes to be transferred to hands; this also helps remove glove powder, which contains latex protein and other glove chemicals that can elicit irritant contact dermatitis or an allergic reaction in sensitized individuals
7. Watches, bracelets, and rings must be removed to prevent harboring of microorganisms; also, rings may perforate glove materials
8. Nails must be kept short and clean and the cuticles well maintained; artificial nails and nail jewelry should not be worn, since research has implicated them in disease transmission in hospitals
9. Intact skin is the best protection against infection; this can be achieved by:
 - a. Minimizing trauma to hands (e.g., cuts, scrapes) while outside the dental setting
 - b. Protecting hands from drying and chapping during cold weather
 - c. Frequent use of lubricating hand lotions
 - (1) When at work, use hand lotion containing petroleum or oil emollients at the end of the day because these ingredients can negatively affect the integrity of latex gloves
10. DHCP who have open or weeping lesions or dermatitis on hands should not provide patient care until the condition resolves, since dermatitis reduces the effectiveness of handwashing, and nonintact skin provides a portal of entry for microorganisms
11. Hand hygiene procedures for routine dental hygiene care include:
 - a. Vigorous lathering of hands using an interlacing finger motion with either antimicrobial or plain soap for at least 20 seconds
 - (1) Plain soap will remove soil and transient microbes
 - (2) Antimicrobial soap will remove or destroy transient microorganisms and reduce resident flora
 - (3) Disposable soap containers should not be refilled, and when empty should be discarded and replaced
 - (4) Soap dishes accumulate microorganisms and are not recommended

(5) Sinks and soap dispensers should be electronically operated or have foot controls; if not, paper towels should be used to turn off sink faucet

b. Rinsing with cool to lukewarm water while rubbing hands together for 10 seconds

c. Drying hands with single-use paper towels

d. Using alcohol-based hand sanitizers as an alternative to handwashing

(1) Used when no visible soil appears on the hands

(2) Have been shown to be effective, sometimes reducing bacterial counts more effectively than soap and water

(3) Most effective at inactivating microorganisms at 60% to 95% alcohol concentrations

(4) May increase the frequency of hand hygiene because of ease of use

(5) According to manufacturer's directions, place an adequate amount of alcohol-based hand sanitizer in hand and then vigorously rub hands until dry

(6) Alcohol-based hand sanitizers may cause dry skin (choose ones that contain emollients)

e. Washing hands between patients, before and after lunch, before and after restroom visits, or any time hands become contaminated

f. Maintaining asepsis by touching only sterile instruments or disinfected surfaces

12. For surgical procedures, use an antimicrobial soap with persistent or residual activity (inhibits growth and survival of microorganisms); chlorhexidine digluconate or triclosan

D. Personal protective equipment (PPE)

1. Protective barriers used to reduce exposure of mucous membranes, hands, and body of DHCP to microorganisms and also to prevent injury from chemicals and particles of debris

2. Used during patient treatment, laboratory procedures, operatory disinfection, and instrument processing

3. PPE includes gloves, masks, protective eyewear, and protective clothing

4. Sequence for donning PPE: protective clothing, then mask and eyewear, and finally, after handwashing, gloves

5. The employer is responsible for providing and maintaining appropriate PPE for employees

6. Gloves

a. The use of gloves provides a high level of protection for both DHCP and patients

(1) Prevents direct contact with microbes in the patient's mouth and on contaminated surfaces (bare hands often will have areas of nonintact skin providing portals of entry for pathogenic microbes)

(2) Prevents saliva and blood from being retained under fingernails; saliva and blood have been shown to persist for several days even with handwashing

(3) Protects against contact with disinfecting and cleaning chemicals and radiograph processing solutions

(4) Protects patients from the microorganisms on the hands of DHCP

(5) Examination gloves provide minimal protection against sharps injuries; utility gloves provide more protection; however, injuries can still occur

b. Risks associated with not routinely wearing gloves

(1) DHCP exposure to potentially infectious patient tissues and contaminated surfaces—most likely route by which DHCP have acquired HBV from patients

(2) Patient exposure to infectious agents originating from the DHCP

(a) Documented source of transmission of HBV from ungloved dentist to patients

- (b) Documented case in which an ungloved hygienist transmitted herpes to 20 of her patients
- (3) Skin irritation from contact with disinfecting chemicals
- (4) Burns resulting from contact with hot items from sterilizer

c. Protocol for glove use

- (1) Wear during intraoral procedures and when in contact with contaminated items or surfaces (e.g., contaminated laundry or waste)
- (2) Exam gloves are single use and disposable (SUD)
 - (a) New gloves are worn for each patient
 - (b) When removing gloves, both pairs should be removed without contamination of hand, wrist, or arm
 - (c) Previously removed gloves should not be reworn but disposed of immediately after removal
 - (d) Removing only one contaminated glove and later regloving will result in contamination of DHCP skin
- (3) If it is necessary to leave the chairside during patient care, remove gloves, and after hand hygiene, don a new pair upon returning (prevents contamination of additional surfaces one may touch and also prevents contamination of the patient with microbes that already may be present on those surfaces)
- (4) Ensure that gloves cover the cuff of a long-sleeved gown
- (5) Change gloves during long appointments because defects in gloves increase with use beyond 60 minutes
- (6) Change gloves when heavily soiled with blood
- (7) Do not wash or disinfect gloves; may cause “wicking” or enhanced penetration of liquids through undetected defects in gloves
- (8) Remove torn or punctured gloves as soon as possible; wash hands and don new gloves
- (9) If using latex gloves, do not use petroleum or oil-based hand lotion or apply petroleum-based lubricants to patient’s lips

d. Types of gloves—Table 10.4 outlines glove materials

- (1) Nonsterile, ambidextrous gloves in sizes extra-small, small, medium, and large are adequate for most procedures; proper glove fit is important to ensure efficient instrumentation and to prevent hand fatigue and possibly carpal tunnel syndrome
- (2) Majority of DHCP are wearing latex-free gloves due to possible DHCP and patient latex allergy
- (3) Sterile gloves are recommended for surgical procedures
- (4) Use FDA-approved puncture-resistant and chemical-resistant utility gloves to prepare chemicals, handle contaminated instruments, and clean and disinfect surfaces
 - (a) Utility gloves are reusable and can be disinfected or sterilized in an autoclave
 - (b) Replace utility gloves when they show any signs of wear (e.g., cracks, punctures, discoloration)
- (5) Overgloves are worn over treatment gloves to prevent cross-contamination of items and surfaces such as pens, charts, and drawers
- (6) Heat-resistant gloves or oven mitts are worn when handling hot items (e.g., unloading sterilizers)

e. Dermatitis and latex allergy

- (1) Irritant contact dermatitis

- (a) Nonimmunologic (nonallergic) reaction of skin to chemicals used in glove manufacturing
- (b) Skin on hands becomes dry, red, itchy, and cracked
- (c) The condition is aggravated by soaps, not rinsing or drying hands completely, perspiration, or cornstarch powder
- (d) Most skin reactions from wearing gloves are caused by irritant contact dermatitis and are not true allergic reactions
- (2) Allergic contact dermatitis
 - (a) Type IV, or delayed, hypersensitivity occurring within hours or days because of allergy to glove chemicals
 - (b) Limited to area of contact, causing itching, redness, and vesicles to appear within 24 to 48 hours, followed by dry skin, fissures, and sores
 - (c) Patch test identifies sensitivity to specific chemical
- (3) Latex allergy
 - (a) Type I, or immediate, hypersensitivity within minutes or hours
 - (b) Allergy to naturally occurring latex proteins
 - (c) Symptoms: skin (hives, swelling, burning, tightness, itching, redness, tingling), lungs (asthma, wheezing, constriction, coughing, sneezing, rhinitis, angioedema), and other (nausea, vomiting, diarrhea, cramps, hypertension, tachycardia, shock)
 - (d) Anaphylactic shock and death can occur with subsequent exposures to latex
 - (e) High-risk individuals for latex allergy include: persons who have had multiple surgeries and persons with spina bifida, urogenital anomalies, spinal cord injuries, and allergies to bananas, kiwis, chestnuts, or avocados
 - (f) Reductions in exposure to latex proteins are known to decrease sensitivity (important for DHCP to reduce their daily exposure to airborne latex proteins by wearing latex-free gloves)
 - (g) The CDC indicates that DHCP need to be educated about skin problems that can occur with frequent hand hygiene and the use of gloves
 - (h) Latex-free environment should be provided to patients and DHCP with a latex allergy
- (4) Procedures for management of persons with latex allergy
 - (a) Include questions in health history appropriate for identifying possible latex allergy
 - (b) Document latex allergy in health history record in a way that will ensure observation by DHCP
 - (c) Schedule allergic patients for the first appointment of the day, when airborne latex proteins are at their lowest levels (still risky)
 - (d) Ensure that DHCP uses latex-free gloves for treatment area preparation and for touching all items that will come in contact with the patient
 - (e) Use latex-free gloves and dental materials during patient care
- 7. Masks
 - a. Purpose
 - (1) Worn to protect the mucous membranes of the nose and mouth from spatter of oral fluids
 - (2) Provide a lesser degree of protection from inhalation of aerosol particles
 - (a) Surgical masks will not provide protection from airborne infections (e.g., TB and severe acute respiratory syndrome [SARS])
 - (b) The N95 respirator is needed to protect against airborne infections, that is, COVID-19
 - (3) May provide some protection to patient from nasal or oral secretions of DHCP

b. Types of exam masks

(1) Dome mask with elastic band

(2) Tie-on or ear-loop mask

c. Composed of synthetic material that should filter at least 95% of small particles

(1) National Institutes for Occupational Safety and Health (NIOSH) certifies three classes of filters (N95%, R-99%, & P-99.97%)

(2) American Society for Testing and Materials (ASTM International) has developed standard specifications for facemask materials in five areas (bacterial filtration efficiency, particulate filtration efficiency, fluid resistance, breathability, and flammability)

(a) ASTM 1—low amounts of fluid exposure

(b) ASTM 2—moderate amounts of fluid exposure

(c) ASTM 3—heavy amounts of fluid exposure (should be used during ultrasonic instrumentation with a face shield when a N95 respirator is unavailable)

(d) N95 respirator used with a face shield for aerosol-generating procedures; using an exam mask over N95 is not recommended because it can alter the fit and allow inhalation of aerosols around the edges; N95 requires medical clearance and fit test before use

d. Maximizing effectiveness and minimizing cross-contamination

(1) N95 should be changed when it becomes moist/soiled; reuse of N95s is only acceptable when there is a shortage of PPE

(2) Change the mask if it becomes moist; moistness compromises its effectiveness by increasing the passage of unfiltered air around the edges of the mask and may wick contaminants through the mask. Refrain from twisting ear loops because this practice will cause gaping of mask and prevent it from fitting snugly against the face

(3) Don a new mask every 20 to 30 minutes to maintain high filterability

(4) Select mask that fits the face well and adjust it to fit snugly against the face, covering the nose and mouth to minimize leakage around the margins

(5) Avoid touching the mask during the appointment

(6) Keep the mask on after completing the procedure to reduce inhalation of aerosols

(7) When removing the mask, handle it by its strings or elastic

(8) Do not leave the mask on the head, dangling around neck, or in your pocket

8. Protective eyewear

a. Purpose—protect the eye from microbial invasion, chemicals, physical projectiles, ultraviolet (UV) light, and lasers

b. Risks for unprotected eyes

(1) Conjunctivitis

(2) Ocular herpes (can cause blindness)

(3) Hepatitis B (eye as portal of entry)

(4) Eye injury (physical or chemical)

c. Types of eyewear

(1) Regular glasses offer limited side or top protection and are not recommended

(2) Safety glasses, goggles, or magnification loupes

(a) Cover entire eye orbit (providing protection on all sides)

(b) Goggles may be worn over prescription glasses

(c) Are more shatter resistant than regular glasses

- (d) Provide minimal visual distortion
- (e) Are able to withstand disinfection (should be cleaned and disinfected between patients)
- (f) Should meet American National Standards Institute (ANSI) guidelines for spatter and impact protection

(3) Face shields

- (a) Worn over eyewear for maximum protection
- (b) Should be chin-length and provide top and side protection
- (c) Masks should still be worn to reduce inhalation of aerosols
- (d) Combination eye shield with attached mask is available
- (e) Provide maximum coverage of face for high-spatter procedures (e.g., air polishing, power scaling)
- (f) Face shields made of thin plastic may have limited impact resistance.

(4) Protective eyewear should be provided to patients

- (a) Protection against damage from UV (curing) light and lasers
- (b) Protection against physical and chemical injury to the patient's eyes during treatment
- (c) Documented case where patient lost an eye due to anesthetic needle penetrating eye and resultant infection

9. Protective (barrier) clothing

a. Purpose

- (1) Protects nonintact skin from contamination by microorganisms
- (2) Reduces the risk of bringing contaminants on unprotected clothing beyond dental setting
- (3) May protect patient from microorganisms on street clothing or may provide protection against disease transmission when soiled protective clothing is changed between patients (fomites—clothing and paper that can absorb and transmit infectious agents)
- (4) Microorganisms adhere to clothing; however, lack of evidence exists to support the extent to which barrier clothing protects against disease transmission

b. Characteristics of protective clothing

- (1) Reusable or disposable. During pandemics or in other situations when the recommendation is to change gowns after each patient, disposable gowns are more convenient
- (2) Gowns, aprons, laboratory coats, or uniforms used as a covering for street clothing or scrub uniforms
- (3) High collar that fits closely around the neck
- (4) Long-sleeved garments with fitted cuffs allow gloves to extend over them for complete coverage
- (5) Type of protective clothing required based on OSHA levels of exposure and procedures being performed

(a) Low level—soiled (laboratory coats)

(b) Medium level—splashed, splattered, or sprayed (fluid-resistant garments)

(c) High level—soaked (fluid-proof garments)

c. To maximize effectiveness and minimize cross-contamination

- (1) Use fabric made of synthetic material, which is more fluid resistant
- (2) Do not wear protective clothing outside treatment area (remove before going out [e.g., to lunch] or leaving the dental setting)
- (3) Change protective clothing daily or when visibly soiled

- (4) Avoid touching clothing during patient care and throughout the day
- (5) Roll protective clothing inside out to minimize contact with exposed surface
- 10. Additional barrier protection used when cleaning or performing surgery
 - a. Plastic aprons or other fluid-proof garments
 - b. Head covers
 - c. Shoe covers
- 11. Laundering of reusable clothing
 - a. Laundered in the office following the manufacturer's instructions
 - b. Sent to a laundry service in a leakproof bag labeled with the biohazard symbol
 - c. It is against OSHA regulations for DHCP to take contaminated protective clothing home for laundering

Table 10.3

Frequency of TB Testing

Risk Classification	Frequency of Testing
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Low	Baseline; then test if TB exposure occurs
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Medium	Baseline, then annually
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Potential ongoing transmission	Baseline, then every 8–10 weeks until evidence of transmission has ceased
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Health care facilities have different TB testing requirements. Facilities should conduct staff TB testing based on risk classification.

From <https://www.cdc.gov/tb>

Table 10.4

Types of Gloves Used in Oral Health Care

Type	Material
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Nonsterile (examination) gloves	
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Latex	
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Nitrile	
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Neoprene	
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Vinyl	
-------	--

Polyurethane	
--------------	--

Styrene-based copolymer	
-------------------------	--

Butadiene methyl methacrylate	
-------------------------------	--

Sterile (surgical) gloves	
---------------------------	--

Latex

Nitrile

Neoprene

Polyurethane

Styrene-based copolymer

Synthetic polyisoprene

Utility gloves

Latex

Nitrile

Neoprene

Butyl rubber

Fluoroelastomer

Polyethylene and ethylene vinyl

Alcohol copolymer

Overgloves

Thin copolymer

Thin plastic (“food handlers”)

Oral Health Care Environment and Promotion of Infection Control

A. Design and equipment selection emphasizes:

1. Smooth construction—eliminates knobs, hooks, and crevices
2. Design of patient chairs and operator stools that:
 - a. Minimize buttons and seams
 - b. Use vinyl upholstery instead of cloth
 - c. Provide foot-control operation
3. Avoidance of fabric-covered, coiled, or mechanically retracted tubings
4. Sink faucets and soap dispensers with foot or electronic controls
5. Paper towel dispenser designed to avoid touching hardware or electronically controlled
6. Plastic-lined waste containers recessed under cabinet, with opening on countertop
7. Surfaces that are compatible with disinfectants and detergents

8. Plastic laminate instead of wood for cabinets and countertops
9. Vinyl flooring and walls that are smooth and seamless
10. Carpeting or wallpaper not recommended
11. Dental unit water lines (DUWLs) that provide:
 - a. Antiretraction valves to prevent the aspiration of microorganisms into water lines
 - b. Equipment, devices, and treatments for DUWLs
 - (1) Filtration unit
 - (2) Sterile-water delivery system
 - (3) Flushing the lines with antimicrobial agents
 - (4) Scheduled testing and monitoring equipment (off-site lab or on-site kit)
 - (5) Periodic chemical shocking to reduce biofilm formation
 - (6) Periodic testing of water quality either on-site or off-site
12. Reduction of airborne microbes
 - a. Air circulation exchange system or single-room filtration units
 - b. Ventilation systems to control noxious sterilization and laboratory vapors
 - c. Prevention of recirculation of contaminated air or transport of microbes
 - d. Cooling and heating system filters that prevent transfer of microbes
 - e. Use of high-volume evacuation systems and devices
13. Housekeeping surfaces are surfaces that are not contaminated by hands or equipment (e.g., walls, floors) and should be cleaned and disinfected routinely or when visibly soiled with detergents or low-level hospital disinfectants
 - a. Clean mops and cloths after use and allow to dry before reuse, or use single-use items
 - b. Periodically clean walls, blinds, and window treatments in patient care areas
 - c. Perform cleaning with a wet cloth or mop to prevent distribution of microorganism-laden dust particles
14. Keep treatment area free of unnecessary or seldom-used equipment and items

Maintaining Asepsis in the Oral Health Care Environment

A. Items associated with oral health care are classified as:

1. Critical—instruments that penetrate oral soft tissue or bone, enter the bloodstream, or enter other sterile tissues of the mouth (e.g., curette); must be heat-sterilized or a SUD
2. Semicritical—items that come in contact with mucous membranes (used in the mouth) but will not penetrate soft tissue, contact bone, enter the bloodstream, or enter other sterile tissues of the mouth (e.g., radiographic positioning devices)
 - a. Preferably heat-sterilized or a SUD
 - b. If heat sensitive, decontaminate using an intermediate level disinfectant with a TB kill claim and/or cover the device with a Food and Drug Administration (FDA)–approved barrier to prevent contamination (e.g., digital radiography sensors)
3. Noncritical—items that contact intact skin (e.g., blood pressure cuff)
 - a. If contamination is possible, cover the device to prevent contamination
 - b. If the device is contaminated, clean and disinfect
 - (1) If contaminated by blood, use intermediate-level disinfectant
 - (2) If not contaminated by blood, use low-level disinfectant
 - c. Noncritical items should be stored after use to prevent contamination

B. Maintaining asepsis with the use of chemicals and surface covers

1. Categories of disinfecting or sterilizing chemicals (Table 10.5)

a. High-level disinfectant—destroys all microorganisms but may not destroy bacterial spores (depends on contact time). Bacterial spore may be destroyed with immersion only depending on contact time. Due to the highly toxic nature, high-level disinfectants should not be used for surface disinfection.

b. Intermediate-level disinfectant—destroys vegetative bacteria, most fungi, and most viruses; inactivates *Mycobacterium tuberculosis* var. *bovis*

c. Low-level disinfectant—destroys vegetative bacteria, some fungi and viruses; does not inactivate *M. tuberculosis* var. *bovis* (Fig. 10.2)

C. Surface disinfection of clinical contact surfaces

1. Any surface that is touched by contaminated hands, instruments, devices, or other items during the provision of oral health care (e.g., light handles, dental equipment)

2. Identify and list surfaces that will be contaminated during treatment

3. Post the list in the treatment area to increase effectiveness and efficiency of decontamination procedures

4. Approaches to clinical contact surface asepsis

a. Avoid unnecessarily contaminating surfaces during treatment by:

(1) Using unit doses of materials to avoid contamination of multiple-use containers

(2) Using prearranged tray setups and cassettes containing a complete selection of instruments needed for a particular procedure

(3) Gathering the needed supplies and equipment at the beginning of treatment to eliminate the need to open drawers and cabinets or to leave the treatment area to retrieve supplies during patient care

(4) Having sterile pliers, overgloves, gauze, and paper towels available for use as barriers between contaminated gloved hands and uncontaminated objects (e.g., drawer handle, papers, floss containers)

b. Prevent contamination by using a surface cover or barrier protection

(1) Surface covers protect surfaces that are difficult to preclean adequately

(2) May be less time-consuming than precleaning or disinfecting surfaces before and after patient care

(3) Surface covers include clear plastic wrap, bags, and tubing

(4) May be placed using clean bare hands on cleaned and disinfected surfaces

(5) Should be placed to protect the entire surface and to ensure that the barrier protection will not come off when the surface is touched

(6) Reduce the use of chemicals that may stain or corrode and are hazardous to DHCP and the environment

c. Disinfection of surfaces not covered by barriers

(1) Requires purchase and proper use of disinfecting chemicals

(2) Chemical use requires that Safety Data Sheets (SDSs) be filed on site and that DHCP be provided information and training on the location, hazards, and proper use of chemicals

(3) Disinfection is best reserved for surfaces that are difficult to cover, smooth surfaces, and those not involving electricity

d. A dual approach consisting of both barrier protection and disinfection with chemicals is most often used in oral health care settings

5. Clinical contact surfaces to be covered or disinfected
 - a. Mobile cart, master control unit, countertop, and radiographic equipment
 - b. Air/water syringe, high-volume and low-volume evacuators (parts of these may be sterilized or disposable, which is preferred because these items come in contact with mucous membranes)
 - c. Hoses, tubing, and controls attached to handpieces
 - d. Supports for handpieces, air/water syringe, and suction devices
 - e. Chairs, including switches and levers
 - f. Control switches
 - g. Light handles
 - h. Faucet handles and soap dispenser (if manually operated)
 - i. Instrument tray (if it cannot be sterilized)
 - j. Stethoscope earpieces
 - k. Floss dispenser, hand mirror, pens, pencils
 - l. Chairside computers and keyboards
 - m. Phones, intercoms
6. Factors influencing effectiveness of surface disinfection with chemicals
 - a. Numbers and types of organisms present; some organisms have a higher resistance to destruction by chemicals (e.g., *Mycobacterium* species)
 - b. Amount of bioburden (blood, saliva, and microorganisms)
 - (1) Organic materials in blood and saliva insulate microorganisms from chemicals
 - (2) May partially inhibit the active ingredient in the disinfectant
 - c. Selection of an Environmental Protection Agency (EPA)–registered intermediate level disinfectant that is tuberculocidal (high-level disinfectant should not be used for clinical contact surfaces)
 - d. Water-based disinfectants are reported to clean organic material better than alcohol-based disinfectants
 - e. Performing procedures carefully and following manufacturer’s instructions by using appropriate disinfectant concentration, use life, and contact time
 - (1) Do not store gauze squares or other materials moistened in disinfectant; instead, wet material before use
 - (2) If using container of disinfectant towelettes, keep lid closed to prevent drying
 - f. Clean surfaces first with a combination of a detergent and a disinfectant to maximize the removal of bioburden before the second disinfection step
 - (1) The detergent or disinfectant used for the cleaning step must contain a surfactant
 - (a) Surfactant—an agent that loosens, emulsifies, and holds soil in suspension, allowing for easier removal of debris and for more thorough cleaning
 - (2) An advantage to using a combination agent is that a chemical disinfectant starts the killing process during the cleaning step and reduces the opportunity for contaminants to spread to other surfaces
7. Cleaning or disinfection technique for clinical contact surfaces
 - a. Exercise care when using disinfectants, and wear PPE, including utility gloves
 - (1) To prevent irritation or injury to eyes, mucous membranes, and skin
 - (2) To avoid breathing in noxious vapors

- b. Step 1—clean; spray the detergent or disinfectant, or apply with a premoistened wipe
- c. Step 2—wipe; vigorously wipe surfaces with a paper towel or premoistened wipe to remove stuck-on debris
- d. Step 3—reapply the disinfectant by spraying or applying with premoistened wipe
 - (1) Leave surfaces undisturbed for a specified contact time outlined by the manufacturer
 - (2) Wipe any remaining wet areas with a paper towel
- 8. Chemical disinfectants for surface disinfection (Table 10.6)—selection criteria
 - a. EPA-registered intermediate-level disinfectant that is tuberculocidal (kills *M. tuberculosis* var. *bovis*)
 - b. Effective within 10 minutes or less
 - c. Does not have an offensive odor
 - d. Reasonable cost
 - e. Provides persistent or residual effect on treated surfaces
 - f. Retains stability and effectiveness in the presence of bioburden, preferably water based
 - g. Good penetrating and cleaning ability
 - h. Compatible with and innocuous to equipment and clinical surfaces (consult product manufacturer for recommendations)
 - i. The CDC does not recommend alcohol, household bleach, or early-generation quaternary ammonium compounds as surface disinfectants

D. Dental unit water line asepsis

- 1. High concentrations of bacteria have been found in untreated DUWLs (e.g., power instrumentation, air-polishing units, high-speed handpieces, air/water syringes)
 - a. Municipal water entering the dental unit is not sterile and contains 0 to 500 colony-forming units per milliliter (CFU/mL) of heterotrophic bacteria
 - b. Water exiting untreated DUWLs may contain more than 100,000 CFU/mL
 - (1) Consists of waterborne bacteria of low pathogenicity or opportunistic pathogens, which pose the greatest risk to immunocompromised individuals
 - (2) Organisms of greatest concern are *Pseudomonas*, *Legionella*, and *Mycobacterium* species
- 2. Microorganisms attach to and accumulate on the inside of the water line tubing, creating biofilm colonies (an organized, protected, and highly resistant colony of live microorganisms attached to a surface) (see the section on “Bacterial Dental Biofilm” in Chapter 14).

Formation of biofilm:

- a. Biofilm includes naturally occurring waterborne bacteria and may include oral microbes from previously treated patients.
- b. Stagnation of water and small-diameter tubing in DUWLs encourages biofilm formation
- c. Incoming water brings a continuous source of nutrients to bacteria
- d. As water flows past the biofilm in the water line, it picks up detached bacteria before exiting through dental equipment
- 3. Dental unit water and infection control
 - a. Recent disease outbreaks prompted a CDC Health Advisory related to public health problems caused by dental unit water
 - (1) In 2011 there was one confirmed case of an 82-year-old woman in Italy who died after contracting Legionnaires’ disease from aspirating *Legionella*-contaminated dental unit water at her dental office

(2) In 2015 and 2016 pediatric patients developed infections after having pulpotomies with DUWLs contaminated with *Mycobacterium abscessus*—a common inhabitant of water, soil, and dust

b. Using water that does not meet the standards for drinking water (potable) increases patient and DHCP exposure to microorganisms (immunocompromised individuals most at risk)

c. CDC recommendations for potable water is no more than 500 CFU/mL of aerobic mesophilic heterotrophic bacteria.

d. Oral health care facility water should meet the CDC standard for routine dental care

e. Dental unit water from a municipal supply should not be used for oral surgery or in the treatment of immunocompromised persons (sterile water delivery systems must be used)

f. Boil water notices in the community

(1) Do not use dental unit or faucet water until the notice is lifted

(2) After the notice is lifted, flush all water lines for 1 to 5 minutes, and disinfect them following manufacturer's recommendations

4. Improving the quality of dental unit water

a. Routinely check antiretraction valves (valves can become stuck in the open position with age)

b. Flush water lines flush for 30 seconds at the beginning of each day and between patients

(1) Flushing does not remove biofilm but may temporarily reduce planktonic (free-floating) microbes

(2) Flushing may help remove patient materials that have entered the turbine, air, and water lines

c. Use an independent water reservoir—a bottle that supplies treated water to the dental unit

d. Chemical treatment of DUWLs is still necessary to control biofilm formation, even when an independent water reservoir is used (in line filters or germicidal tablets)

e. Periodically shock DUWL to reduce biofilm formation

f. Test the quality of water periodically as needed to ensure compliance

g. Keep records of shocking and testing results

Table 10.5

Categories of Disinfecting or Sterilizing Chemicals

Category	Definition	Examples	Use
High-level disinfectant	Destroys all microorganisms, but not necessarily high numbers of bacterial spores	Glutaraldehyde, glutaraldehydephenate, hydrogen peroxide, hydrogen peroxide with peracetic acid, peracetic acid, orthophthaldehyde	Heat-sensitive reusable items: immersion only. Rarely used due to highly toxic nature
Intermediate-level disinfectant	Destroys vegetative bacteria, most fungi, and most viruses; inactivates <i>Mycobacterium tuberculosis</i> var. <i>bovis</i> (is tuberculocidal)	Environmental Protection Agency (EPA)—registered hospital disinfectant with label claim of tuberculocidal activity (e.g., chlorine-based products, phenolics, iodophors, quaternary ammonium compounds with alcohol, bromides)	Clinical contact surfaces, noncritical surfaces with visible blood
Low-level disinfectant	Destroys vegetative bacteria, some fungi, and some viruses; does not inactivate <i>M. tuberculosis</i> var. <i>bovis</i> (is not tuberculocidal)	EPA-registered hospital disinfectant with no label claim of tuberculocidal activity (e.g., quaternary ammonium compounds)	

Housekeeping surfaces (e.g., floors, walls); noncritical surfaces without visible blood; clinical contact surfaces

aSome, but not all, of these products can serve as high-level disinfectants and sterilants, depending on the immersion time used.

bA hospital disinfectant is one that has been shown to kill *Staphylococcus aureus*, *Pseudomonas aeruginosa*, and *Salmonella choleraesuis*.

cThe Centers for Disease Control and Prevention (CDC) indicates that low-level disinfectants can be used on clinical contact surfaces if the product has a label claim of killing human immunodeficiency virus and hepatitis B virus in addition to being an EPA-registered hospital disinfectant.

Modified from Centers for Disease Control and Prevention. Guidelines for infection control in dental health-care settings. MMWR. 2003;52(RR-17):1–66.

FIG. 10.2Decreasing order of resistance of microorganisms to germicidal chemicals. Modified from Centers for Disease Control and Prevention. Guidelines for infection control in dental health care settings. MMWR. 2003;52(RR-17):1–68.

Follow for extended description

Table 10.6

Chemical Agents for Surface Disinfection

Chemical Classification	Advantages	Disadvantages
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Chlorines		
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Rapid acting		
--------------	--	--

Broad spectrum		
----------------	--	--

Economical		
------------	--	--

Prepare solution daily		
------------------------	--	--

Diminished activity by organic matter		
---------------------------------------	--	--

Corrosive		
-----------	--	--

Strong odor		
-------------	--	--

Iodophors		
-----------	--	--

Broad spectrum		
----------------	--	--

Few reactions		
---------------	--	--

Residual biocidal activity

Unstable at high temperatures

Dilution and contact time critical

Prepare solution daily

Discoloration of some surfaces

Inactivated by hard water

Synthetic phenolics

Broad spectrum

Residual biocidal activity

Compatible with most metals

Degrades certain types of plastic over time

Difficult to rinse

Film accumulation

Alcohol-based products are only fair to poor in cleaning ability

Dual or synergized quaternaries

Broad spectrum

Contains detergent for cleaning

Easily inactivated by anionic detergents and organic matter

Damaging to some materials

Oxidizers Broad spectrum, Fast acting, Breaks down to nonhazardous components

Corrosive to some materials

Maintaining the Treatment Area During Patient Care

A. Spatter and aerosol management to reduce the number of microorganisms escaping from the source

1. Use a preprocedural, antimicrobial rinse containing chlorhexidine, diluted hydrogen peroxide, essential oils, or iodophor to reduce microbial counts in spatter and aerosols

2. Use air and water separately instead of a combination spray
3. High-volume evacuation (HVE) is strongly recommended for aerosol-generating procedures.
A wide variety of HVE systems are available

a. When using the saliva ejector without a one-way valve, do not allow the patient to close the lips around the tip since previously suctioned fluids might enter a patient's mouth by reverse flow in the vacuum line

b. Use one-way valve attached to saliva ejector to prevent backflow of oral fluids

4. Use a rubber dam to reduce spatter and aerosols and to lessen the chance of a patient's saliva retracting into the dental handpiece and the air/water syringe

B. Reducing the risk of contamination and disease transmission

1. Limit the areas of contamination; use overgloves when touching clinical contact surfaces

2. Disinfect anything not covered with barriers when touched by contaminated hands

3. Avoid accessing drawers, cabinets, and other storage areas with contaminated gloves; ask for assistance if additional supplies are needed

4. Protect paper and electronics from contamination

a. Ask for the help of hygiene assistant or other personnel to record

b. Use overgloves

c. Use audio or computer programs to record data if making manual entries

d. Generate patient record with voice-activated computer software

5. Eating, drinking, smoking, applying of cosmetics or lip balm, and handling of contact lenses are prohibited in patient care areas

6. Food and drink cannot be stored in refrigerators, freezers, or cabinets or on shelves and countertops where blood or saliva is present

C. Waste disposal during treatment

1. Immediately discard blood-soaked items in a small biohazard bag taped to the mobile cart or the cabinet

2. Do not allow contaminated waste to accumulate on trays (e.g., exam gloves)

D. Precautions and care in handling syringes or sharp instruments

1. Needlestick injuries are a major cause of disease transmission to health care personnel

a. Never permit "sharps" to be directed toward the body

b. Do not allow uncovered needles to remain on tray

c. Never recap a needle using a two-handed technique

(1) Using one hand, the cap is scooped up from the tray

(2) Use the sheath holder

(3) Use self-sheathing disposable needles

d. Never bend, break, or otherwise manipulate a needle

e. DHCP giving an injection should appropriately recap the needle to eliminate the danger of passing an uncovered needle to another worker for recapping

f. Sharps containers should be located in each treatment area

2. Instrument sharpening

a. Sharpening contaminated instruments poses risk of disease transmission

(1) Avoid sharpening during a procedure

(2) Instead, include duplicates of most-used instruments in the cassette for difficult cases

b. The ideal method involves sharpening sterile instruments and then sterilizing again before use

3. Avoid injury with contaminated instruments during patient care (i.e., curettes and scalers)

a. Avoid fulcruming on the same tooth being treated

b. Never wipe instruments on gauze held in your hand or wrapped around your finger

c. Instruments should be organized and lying flat (instruments should not be on top of each other, sticking up, or balancing crosswise on the edges of cassettes)

d. Never grab instruments by the working end

e. Avoid holding more than one instrument in each hand

f. Do not adjust dental light while holding an instrument

E. Exposure incident protocol (Box 10.2)

1. Includes all needlesticks, puncture wounds, cuts, and scrapes with contaminated instruments and all nonintact skin and mucous membrane exposure to blood and saliva

a. In DHCP 90% of exposures result from sharps penetrating the hands and fingers

2. Immediately wash the injured area with antimicrobial soap and water or flush with water for a mucous membrane exposure (eye)

3. Inform the employer about the incident

4. If the source individual can be identified, request their consent for blood testing for HBV, HCV, and HIV as soon as possible (if the source individual's disease status is not already known)

5. Results of the source individual's tests are confidential and are revealed only to the exposed employee, not to the employer

6. The employer is responsible for providing to the exposed employee and patient laboratory testing for HBV, HCV, and HIV (30-minute rapid test available), medical evaluation, and counseling by a preselected, qualified, licensed health care provider (HCP)

7. HCP should be familiar with most current United States Public Health Service recommendations for testing and postexposure prophylaxis (PEP)

8. Within 2 hours of exposure, the employee should seek medical care allowing PEP, if indicated, which offers the greatest chance of success in preventing disease transmission (seroconversion)

a. Seroconversion depends on dose of blood transferred, titer of virus (viral load), resistant viral strain, depth of injury, and host factors

b. PEP may reduce the risk of infection by 80%

9. If the employee declines testing, then a blood sample from the employee may be preserved for 90 days in case the employee later consents to testing

10. The HCP informs the employee about both their own test results and the source patient's test results and of any conditions that require further evaluation and treatment

11. The employer receives a written report from the HCP confirming that the employee was tested, informed of the results, and counseled if further evaluation or treatment was needed

a. Report notes if hepatitis B vaccine was administered

b. Report does not include test results, diagnoses, or treatment because this information is confidential and protected employee health information

F. Prevention of disease transmission during radiographic procedures

1. Although radiographic procedures are generally noninvasive, the potential for disease transmission does exist

2. The use of barriers on equipment will prevent contamination and promote efficiency
 - a. Plastic bag to cover position-indicating device (PID), tubehead, and swivel arms
 - b. Plastic cover for exposure control switch
 - c. Cover for headrest and chair controls
 - d. FDA-approved plastic barrier for digital sensor
 - e. Plastic barrier underneath exposed receptors and holders and on countertop
3. If contamination occurs, disinfect
 - a. Use EPA-registered intermediate-level disinfectant
 - b. Disinfect all touched surfaces that are unprotected by barriers
 - c. Do not directly spray disinfectant on the control panel because it has electrical components
 - d. Follow manufacturer's directions for digital sensors (most cannot be disinfected)
4. Radiographic receptor holders—use reusable holders that can be heat-sterilized, or use disposable items (SUDs)
5. Radiographic receptor options
 - a. Digital radiographs (sensor—immediate image with no processing necessary; phosphor plates (PSP) require nonchemical processing)
 - b. Film in plastic covering or pouch (best option when using film)
 - c. Film without plastic covering (during processing, special handling is necessary when removing the film from the packet to avoid contaminating the film)
 - d. Film is being used less frequently
6. Aseptic procedure for imaging and processing dental radiographs in plastic pouches or film
 - a. Wear PPE
 - b. After exposing the film, drop it into a disposable cup
 - c. Remove the plastic pouches and drop the film packet into a new cup without contaminating the film packet or the outside of the cup
 - d. Remove gloves and wash hands before transporting the film to the processing area
 - e. With ungloved hands, unwrap the film and put it into the automatic processor
7. Aseptic procedure for processing dental radiographs without plastic covering
 - a. After exposure, contaminated film should be placed in a plastic cup without contaminating the outside of the cup
 - b. Don a new pair of gloves before transporting the film to the darkroom
 - c. Open the film packet and drop the film on a paper towel (proceed carefully to avoid contaminating the film)
 - d. Remove gloves, wash hands, and with bare hands, feed the film into the automatic processor
8. Aseptic technique for processing phosphor plate covered with disposable barrier
 - a. After exposure, contaminated plates should be placed in a cup
 - b. Don a new pair of gloves before transporting the cup to PSP processor
 - c. Tear disposable barrier and squeeze the plate into an open PSP box without contaminating the inside of the box
 - d. Dispose of contaminated barriers and gloves
 - e. Place plates into the processor
 - f. After plates have exited processor, don gloves and replace barriers
- G. Infection control for the dental laboratory
 1. Prevention of disease transmission between the treatment area and the dental laboratory

2. OSHA regulations include measures for the protection of dental laboratory personnel from bloodborne pathogens
3. Impression materials, prostheses, and appliances are exposed to oral microflora
4. Protective attire and barrier techniques
 - a. When performing laboratory procedures, PPE should be worn; use caution with lathes because gloves can become caught and cause injury
 - b. Masks and eyewear protect against chemicals, aerosols, spatter, and projectiles
5. Preparation of materials and transport to laboratory
 - a. Clean and disinfect impressions, prostheses, and casts before transport
 - b. Always disinfect prostheses before sending them to another location or returning to the patient
 - c. Use new packing material every time the prosthesis is transported between the laboratory and the oral health care facility to prevent contamination of shipping materials
 - d. Communicate the infection control protocol to the laboratory and DHCP
6. Minimizing contamination of common areas in laboratory
 - a. Use paper covers on countertops
 - b. Place plastic barriers on frequently touched areas
 - c. Procedures for using the lathe
 - (1) Use fresh pumice each time
 - (2) Use disposable trays
 - (3) Use sterile or disposable ragwheels (polishing device made of cloth)
7. Use a shielding device and air-suction motor with the polishing lathe, model trimmer, or grinding bench to minimize aerosol spray
8. An EPA-registered intermediate-level disinfectant should be used to disinfect contaminated laboratory materials (e.g., impressions, fixed and removable prostheses, retainers, crowns)
 - a. Spray the chemical or, preferably, immerse the material in the chemical using minimal effective exposure time (at least 15 minutes) to prevent damage to the materials
 - b. Consult with the manufacturer to determine which chemicals are compatible with the item to be disinfected and the immersion time required
9. Professional cleaning of removable prostheses
 - a. Use of denture-cleaning solutions is not a substitute for disinfection
 - b. To prevent microbial contamination:
 - (1) Use resealable plastic bags
 - (2) Do not reuse the denture-cleaning solution or plastic bags
 - (3) Place sealed bags in a disinfected beaker in the ultrasonic unit
 - (4) Sterilize or dispose of the equipment used in cleaning (e.g., brushes)
 - c. Rinse the prosthesis with water to remove any residual chemical and place it in a mouthwash solution to ensure a pleasant taste for the patient

Box 10.2

Recommendations for Contents of the Occupational Exposure Report

Date and time of exposure

Details of the procedure being performed, including when and how the exposure occurred

If related to a sharps device, the type and brand of device and how and when in the course of handling the device the exposure occurred

Details of the exposure, including the type and amount of fluid or material and the severity of exposure; for example:

For a percutaneous exposure, depth of injury and whether fluid was injected

For a skin or mucous membrane exposure, the estimated volume of material and the condition of the skin (e.g., chapped, abraded, intact)

Details about the exposure source; for example:

Whether the source material contained hepatitis B virus (HBV), hepatitis C virus (HCV), or human immunodeficiency virus (HIV)

If the source is HIV-infected, the stage of disease, history of antiretroviral therapy, viral load, and antiretroviral resistance information, if known

Details about the exposed person (e.g., hepatitis B vaccination and vaccine response status)

Details about counseling, postexposure management, and follow-up

Maintaining the Oral Health Care Environment After Patient Care

A. Decontamination of the treatment area

1. Wear PPE, including utility gloves
2. Seal and transport the biohazard bag (that is taped to the mobile cart-if applicable) containing blood-soaked gauze to the biohazard waste container
3. Deliver contaminated instruments and cassettes to the instrument-processing area on a tray or in a puncture-resistant container
4. Remove surface barriers carefully to avoid contamination of the surfaces underneath, and dispose of in a trash receptacle
5. Remove and dispose of SUDs (air/water syringe tip, saliva ejector)
6. Using an air/water syringe, flush low-volume and high-volume (if used) evacuator tubing for 30 seconds
7. Flush all other water lines for 30 seconds between patients (ultrasonic unit)
8. If instrument sterilization is delayed, place the instruments in a holding solution to prevent bioburden from drying on the instruments
9. At the end of the workday, flush high-volume and low-volume evacuator tubing with a cleaning and disinfecting solution; at least weekly, clean the trap of the evacuation system or, preferably, replace with a disposable trap

B. Regulated infectious waste

1. Medical waste that has the potential for disease transmission and requires special handling and disposal
2. OSHA regulates the handling of infectious waste; the EPA regulates the disposal of infectious waste for three categories:
 - a. Sharps (e.g., needles, scalpel blades, instruments)
 - b. Tissue and extracted teeth; CDC allows extracted teeth to be returned to the patient
 - c. Blood, items soaked or caked with blood, or items that could release potentially infectious materials when compressed; CDC considers saliva as infectious material but not as regulated medical waste
3. Unregulated medical waste—solid waste that is generated, including all disposable items other than sharps, tissues, extracted teeth, and blood-soaked items (e.g., gloves, saliva

ejectors, cups, surface barriers); unregulated medical waste may be potentially infectious; however, it is unregulated and may be disposed of as common trash unless restricted by state and local laws

4. Contact state and local governments for regulations on the disposal of regulated and unregulated medical waste

C. Waste management

1. Line trash receptacles with plastic bags

2. Sharps containers should be placed in each treatment area

D. Waste disposal

1. Regulated infectious waste must be separated from other waste, put in a color-coded (“red bagged”) container, and labeled with the biohazard symbol (Fig. 10.3)

a. Sharps must be in a closed, leakproof, and puncture-resistant sharps container

b. Nonsharp items must be placed in a leakproof container (e.g., biohazard bag)

2. Disposal options

a. Sterilization on-site and disposal according to local and state regulations

(1) Do not process instruments and waste in the same load

(2) Properly perform biologic monitoring during sterilization

(3) Extracted teeth, to be used for educational purposes, with amalgam restorations must not be heat-sterilized because this will generate toxic vapors; use a chemical sterilant instead; teeth without restorations must be heat sterilized

b. Hazardous waste removal service by an EPA-approved waste hauler

(1) The oral health care facility is ultimately responsible for the proper disposal of infectious waste; therefore the waste hauler’s credentials should be checked carefully

(2) A manifest should be provided by the waste hauler indicating the method by which the waste was treated and its final disposal site

E. Instrument processing

1. Organization of sterilization area

a. For efficiency, the instrument-processing area should be centrally located

b. The area is designed to allow for linear progression of instrument processing and for the separation of contaminated and clean or sterile items

c. The area should be divided into three designated areas:

(1) Decontamination area—receiving area for contaminated instruments (soaking and waste disposal), ultrasonic cleaning, rinsing, and drying

(2) Packaging area—instruments are arranged in sets; chemical and biologic indicators (BIs) are added; instruments are wrapped or bagged, sealed, and labeled

(3) Sterilization and storage area—instruments are processed and stored

2. Decontamination area procedures

a. Use PPE, including utility gloves

b. If a delay occurs before decontamination, soak instruments in a holding solution to prevent the drying of saliva and blood

(1) Extended soaking is not recommended because it can damage the instruments

(2) The holding solution can be water, enzyme solution, heparin solution (dissolves blood), or detergent

c. Cleaning instruments in an ultrasonic unit or instrument washer is an important step; otherwise, stuck-on blood or other organic material will remain on the instruments even after sterilization (instruments with adhered debris are not sterile)

d. Ultrasonic cleaning or instrument washer or washer/disinfector

(1) Ultrasonic cleaning uses vibrations dislodge and dissolve organic material

(2) Instrument washers or washer/disinfectors use water pressure and detergents to dislodge and dissolve organic material

(3) The safest and most efficacious method of cleaning

(4) Eliminates the need for hand scrubbing and potential for injury with contaminated instruments

(5) Procedure for using ultrasonic cleaners and instrument washers

(a) Use solution(s) specifically designed for the unit; may contain antimicrobial properties

(b) Cycle time depending on the unit's instructions for use, the instruments, and the amount of material on the instruments (instruments in cassettes might require longer exposure time)

(c) Keep the unit covered to prevent exposure to aerosols

(d) Rinse the instruments and let excess water drain from the cassettes

(e) Open the cassette and inspect instruments to ensure absence of debris

(f) Drain ultrasonic solution, disinfect unit, and replace with new solution daily

(g) Periodically test the unit (aluminum foil test for ultrasonic cleaners; soil test for instrument washers)

e. Manual scrubbing technique—used rarely when bioburden remains on instruments or when ultrasonic exposure would cause damage to instruments or devices (some handpieces)

(1) Scrub with a long-handled, stiff brush; wear utility gloves

(2) Scrub the item, holding it low in the sink to minimize spatter

(3) Manual scrubbing is not advised because it poses an increased risk for injury

f. Asepsis for high-speed and low-speed dental handpieces (anything that is connected to a dental air or water line)

(1) Refer to the manufacturer's instructions for cleaning, lubrication, and sterilization

(2) Some handpieces can be cleaned ultrasonically; all handpieces must be able to withstand heat sterilization and must be sterilized before reuse

(3) Flush the water lines of high-speed handpieces for 30 seconds after use

g. Prophylaxis angles; should be heat-sterilized or SUDs; consult the manufacturer's instructions for cleaning, lubrication, and sterilization

3. Packaging area procedures

a. Surgical and hinged instruments

(1) Instruments can be dipped in surgical milk to prevent rusting and to lubricate hinged instruments (rusted instruments should be discarded and never used because sterility of these instruments cannot be ensured)

(2) Hinged instruments should be opened before packaging

b. The instruments to be bagged should be dry to avoid penetrating the packaging, and excess moisture can interfere with sterilization

c. Packaging materials vary according to the type of sterilization method (Table 10.7)

d. Packaging materials include plastic or paper pouches, plastic tubing, cassettes, or trays to be wrapped with paper or cloth, sealed with appropriate tape, dated, and labeled with the date, type of instrument setup, and sterilizer used (in the case of more than one sterilizer)

(1) With more than one sterilizer in use, if one was to malfunction, it would be important to identify the packages that had been processed in that sterilizer

e. To maintain package integrity, do not seal the package using staples, pins, or paper clips

f. Packages should be folded on the perforation (autoclave bags) and tightly sealed to eliminate gaps

g. Autoclave bags are designed for single use because of color change chemical indicators and pores in the paper that open during sterilization, allowing steam to penetrate, and then close on cooling to inhibit microbes from entering into package

h. The packaging materials used must NOT:

(1) Melt or char

(2) Prevent the sterilizing agent from penetrating

(3) Be easily torn by sharp instruments

i. Instruments should be packaged loosely to allow for maximum contact with steam or chemical vapors; overloaded packages can lead to incomplete sterilization

j. Biologic and chemical indicators are added during packaging procedures

4. Sterilization area procedures

5. Sterilization methods (see Table 10.7)

a. Steam (moist heat under pressure) sterilization—autoclave

(1) Types based on how air is removed from the chamber

(a) Gravity displacement

(b) Vacuum pump (type B sterilizer)—perform air removal test (Bowie-Dick) at the beginning of each day to ensure complete air removal from the chamber

(c) Pressure pulse

(2) Moist heat denatures and coagulates microbial proteins; pressure serves to elevate temperature

(3) Periodically check gasket on door to ensure a tight fit

(4) Steam must be able to penetrate the instrument package; pack loosely to allow for free passage of steam

(5) Packages and cassettes should be placed on their edges, not laid flat

(6) Avoid the package coming in contact with chamber walls

(7) Use distilled or deionized water to prevent deposits on instruments

b. Dry-heat sterilization

(1) Sterilizes by oxidizing cell parts

(2) Keeps air spaces between packages

(3) If the sterilizer is opened, timer must be restarted

c. Unsaturated chemical vapor sterilization

(1) Alcohols, formaldehyde, ketone, acetone, and water are heated under pressure to produce the gas that permeates and destroys microbes

d. Ethylene oxide gas sterilization

(1) Used mostly in hospitals and large clinics

(2) Operation

- (a) Sterilize at room temperature, 75°F (25°C), for 10 to 16 hours
 - (b) Alternatively, sterilize at 120°F (49°C) for 2 to 3 hours
 - (c) Aerate plastic and rubber materials for at least 16 hours after sterilization to remove gas molecules
 - (d) Warning: residual gas can cause tissue burns
 - e. Liquid chemical sterilants
 - (1) Used for heat-sensitive, semicritical items when ethylene oxide gas is not available
 - (a) Should not be used as a substitute for heat sterilization of semicritical items that are not heat sensitive
 - (b) Should be avoided because mostly heat tolerant, or SUDs are available
 - (c) Biologic monitoring is not possible, so this method cannot ensure sterility
 - (2) Follow the manufacturer's instructions for dilution, temperature, and contact time
 - (3) Ensure that the instruments are dry to avoid diluting the solution
 - (4) Remove the instruments with sterile forceps, rinse with sterile water, and package or cover with sterile towel
 - (5) Chemicals used for sterilization (see Table 10.6); the disadvantages of chemicals are:
 - (a) They produce toxic fumes and are irritating to tissues
 - (b) BIs are not available with the use of chemical sterilants; therefore, the sterility of items cannot be verified
 - (c) When removed from chemicals, items are exposed to the environment and can easily become contaminated because of lack of packaging
 - (d) Chemical sterilization requires an extended contact time of 3 to 12 hours, depending on the chemical
 - 6. Verification of sterilization
 - a. Failure of sterilization can occur as a result of operator or mechanical failure
 - (1) Overloading—the reason for failure in the majority of cases
 - (2) Improper packaging
 - (3) Improper timing
 - (4) Improper unit operation
 - (5) Unit malfunction
 - (6) Improper maintenance of equipment
 - b. The CDC recommends routine use of BIs (spore tests) for verification of sterility
 - (1) Biologic monitoring is a process in which highly resistant spores are passed through the sterilizer and then cultured to determine whether they have been killed
 - (a) If spores have been killed, all the less-resistant microorganisms exposed to the same conditions will also have been destroyed
 - (b) Primary method to ascertain sterility
 - (c) Biologic monitoring with BIs can be done in the dental setting or in a processing facility
 - (2) Types of BIs
 - (a) Spore strips—paper strips containing one or two types of spores enclosed in a glassine envelope
- [1] After sterilization, remove the spore strip aseptically, place it in a culture medium, and incubate for 7 days

[2] If spores are present, they will grow and change the color of the growth medium, indicating sterilization failure

[3] Strips can be used for all methods of heat sterilization

(b) Self-contained vial—contains spore strip with an ampule of growth medium in a plastic vial

[1] After sterilization, squeeze the vial to break the internal ampule, which would mix the growth medium with the spores

[2] Incubate the vial; if spores are present, they will multiply and change the color of the growth medium, indicating sterilization failure

[3] Vials can only be used in a steam autoclave

(c) Specific bacterial endospores used in BIs

[1] Steam autoclave and chemical vapor sterilizer—spores of *Geobacillus stearothermophilus*

[2] Dry heat and ethylene oxide gas sterilizer—spores of *Bacillus atrophaeus*

(3) Use of BIs

(a) Place a strip inside one of each type of package (e.g., autoclave bag, cassette)

(b) Incubate a control BI that has not been sterilized, along with the test BI

(c) Growth of spores from the control BI confirms that live spores were present

(d) No growth from the test BI indicates that sterilization has been achieved

(4) Use BIs once a week (minimally) to test sterilization equipment and when:

(a) The equipment has been repaired

(b) New packaging material is used

(c) A new sterilizer is operated

(d) Training new staff

(e) The loading procedure is changed

(f) Sterilizing an item to be implanted

(5) Sterilization failure (growth on the BI test, or positive spore test)

(a) Review the procedures to determine any operator error

(b) Take the sterilizer out of operation and retest with mechanical, chemical, and biologic monitors

(c) If the repeat BI test is negative, put the sterilizer back in service

(d) If the repeat BI test is positive, determine the cause of failure, and repeat the BI test three times before putting the sterilizer back into service

(e) Withdraw the instruments and repeat the process

(6) Documentation

(a) Record the results of biologic monitoring in a logbook

(b) Necessary for compliance with federal regulations

(c) Serves quality assurance and risk management purposes

c. Chemical monitoring

(1) Chemical indicators—items containing heat-sensitive chemicals that change color when exposed to certain temperatures to assess conditions during the sterilization process (e.g., autoclave tape, special markings on autoclave bags, chemical indicator strips)

(2) Chemical monitoring does not provide proof of sterilization

(3) Six Types of chemical indicators

a) Class I (process indicators)—external chemical indicator changes color after a certain temperature is reached (e.g., autoclave tape)

1. Distinguishes those instruments that have been in the sterilizer from those that have not, preventing accidental use of unprocessed items
2. Should be present on packaging material or applied on the outside of every instrument package and cassette
 - b) Class II (Indicators for Specific Tests)—air removal test for autoclaves (e.g., Bowie-Dick test)
 - c) Class III (Single Parameter)—pellets, not used in dentistry
 - d) Class IV (Multivariable)—have internal and external indicators (e.g., paper/plastic peel pouches)
 - e) Class V (Integrating)—changes color slowly when exposed to a combination of time, temperature, and steam; placed inside each instrument package to confirm whether instruments have been exposed to sterilizing conditions (e.g., indicator strip placed inside a cassette)
 - f) Class 6 (Emulating)—are specific tests similar to a class V designed to react to all critical variables of a specified sterilization cycle (e.g., used in a prevacuum, ex-press cycle). Not a substitute for a biological indicator.
- d. Mechanical monitoring
 - (1) Observation of sterilizer gauges, including temperature, pressure, and exposure time
 - (2) Incorrect reading indicates a functional problem
7. Storage area procedures
 - a. After sterilization, allow the instrument packages to cool and dry before storage
 - (1) Microorganisms and instrument tips can penetrate wet packaging material and compromise sterility
 - (2) Microorganisms on contaminated surfaces can wick through wet packaging
 - b. Instruments must not be stored unwrapped or unpackaged (unpackaged instruments are immediately contaminated when exposed to the environment)
 - c. Packaged instruments should not be stored above the sterilizers and should be stored away from treatment areas to lessen chances of contamination
 - d. Sealed packages should be kept on shelves protected by doors for dry, low-dust storage
 - e. Sealed instrument packages have been shown to maintain sterility for up to 6 months; however, to err on the side of caution, repackage and sterilize after 30 days
 - f. Instrument packages should be rotated so that the oldest dated instrument pack is used first
 - g. Packages that are dropped on the floor, punctured, torn, or wet are considered contaminated and must be reprocessed
 - h. Wrapped cassettes are ideal for storing and then serving as sterile instrument trays

FIG. 10.3 Biohazard symbol. From Kelly R. Mosby's Textbook for Long-Term Care Nursing Assistants. 7th ed. Elsevier; 2014.

Table 10.7

Comparison of Heat Sterilization Methods

Method	Sterilizing Conditions ^a	Packaging Materials	Advantages	Precautions
Steam Autoclave				
Paper wrap				

Nylon "plastic" tubing

Time-efficient

Good penetration

No closed containers

Damage to plastic and rubber

Standard cycles 20–30 min at 250°F

Paper or plastic peel pouches

Thin cloth

Wrapped, perforated cassettes

Good penetration

Accurate chemical indicators (classes 4 and 5)

Non–stainless steel items corrode

Use of hard water may leave deposits

Items may be wet after cycle

Flash cycles 3–10 min at 273°F No packaging (unwrapped) Quick turn-around

Unwrapped items quickly contaminated after processing

Unsaturated chemical vapor 20 min at 270°F

Paper wrap

Paper or plastic peel pouches

Wrapped, perforated cassettes

Time-efficient

No corrosion

Items dry quickly after cycle

No closed containers

Damage to plastic and rubber

Must use special solution

Predry instruments

Provide adequate ventilation

Cannot sterilize liquids

No cloth wrap (absorbs too much chemical vapor)

Dry heat 60–120 min at 320°F Paper wrap No corrosion
Some paper may get charred

Predry instruments

Oven type (static-air)
Appropriate nylon “plastic” tubing

Closed containers

Wrapped, perforated cassettes

Can use closed containersb

Low-cost

Items are dry after cycle

Long sterilization time (oven type)

Damage to plastic and rubber

Door cannot be opened during cycle

Cannot sterilize liquids

Rapid heat transfer (forced air)
12 min at 375°F (wrapped)

6 min at 375°F (unwrapped)

Aluminum foilc
No corrosion

Short cycle

Items are dry after cycle

Unwrapped items quickly contaminated after processing

a These conditions do not include warm-up or cool-down time, and they may vary, depending on the contents and volume of the load and brand of the sterilizer.

b Biologic indicator must be used to ensure sterility.

c Tears or punctures easily.

Modified from Miller CH. Infection Control and the Management of Hazardous Materials for the Dental Team. 7th ed. Elsevier; 2022.

Governmental Agencies and Infection Control

A. FDA

1. Regulates all medical devices

a. Examples in oral health care include biologic and chemical indicators, ultrasonic cleaners, PPE, dental instruments, and sterilizers

2. Ensures products are safe and effective

3. Ensure devices fulfill performance standards before marketing

B. EPA

1. Mission to protect human health and the environment

2. Ensures the safety and effectiveness of chemical disinfectants (EPA registration number displayed on container)

3. Regulates medical and chemical waste after leaving the oral health care (OHC) setting

C. CDC

1. Under the auspices of the US Department of Health and Human Services

2. Health and quality-of-life promotion mitigating injury, disability, and disease

3. Infection control guidelines and recommendations for dentistry (infection procedures used in oral health care settings are based on 2003 recommendations and 2016 enhancements)

4. Advisory agency only; does not have authority to enforce guidelines unless language is adopted into a state practice, which then becomes regulatory

5. In 2016 the CDC formalized the recommendation that all dental settings should designate an infection control coordinator (ICC). Assign at least one individual trained in infection prevention responsibility for coordinating the program.

6. The ICC should be responsible for developing written infection prevention policies and procedures, known as Standard Operating Procedures (SOP) based on evidence-based guidelines, regulations, or standards.

7. Policies and procedures should be tailored to the dental setting and reassessed on a regular basis (e.g., annually)

D. OSHA

1. Within the US Department of Labor (federal and state divisions)
2. Ensuring a safe and healthy working environment for employees
3. OSHA formulates and has the authority to enforce regulations related to occupational safety and health
4. Protection of DHCP against exposure to bloodborne pathogens, safety with chemicals, other contagions, and other workplace hazards
5. COVID-19 pandemic guidance
 - a. Inform DHCP about their risks of exposure to SARS-CoV-2, what employer site-specific measures enacted to protect them, and how they can protect themselves
 - b. Recommendations for administrative, work, and engineering controls following CDC guidelines
 - c. Recommendations for return-to-work guidance
6. OSHA Bloodborne Pathogens Standard
 - a. The most important infection control regulation in dentistry for the protection of DHCP
 - b. The final Bloodborne Pathogens Standard was published in 1991 and became effective in 1992
 - c. The Needlestick Safety and Prevention Act was added to the Standard in 2001
 - (1) On an annual basis, employers must solicit input from DHCP to identify, evaluate, and select safer medical devices to minimize or eliminate occupational exposures (e.g., self-sheathing needles)
 - d. Employers are responsible for staff compliance with the Standard; this includes:
 - (1) Review of the Standard and ensuring that a copy is available on-site
 - (2) Formulation of a written exposure control plan that contains:
 - a) Clarification of which employees face a potential risk for occupational exposure and will be covered under the Standard
 - b) Description of how and when provisions will be implemented (i.e., communication of the hazards to employees, hepatitis B vaccination, postexposure evaluation and follow-up, record keeping, use of PPE, engineering and work practice controls, and housekeeping)
 - c) Prevention and evaluation of exposure incidents (sharps injuries)
 - d) In 2005 OSHA determined that oral health care settings are of low hazard and are exempt from keeping an injury and illness log
 - e) Employers still must report any workplace incident that results in the hospitalization of three or more employees or a fatality
 - (3) Training of employees
 - a) Training that provides information about the hazards and preventive measures related to occupational exposure to bloodborne pathogens
 - b) Training to be completed at the initial time of assignment and annually thereafter
 - c) Person conducting the training must be qualified and knowledgeable
 - d) Training must provide an opportunity for interactive questions and answers with the person conducting the training
 - (4) Providing employees with necessary materials to comply with the Standard
 - a) Offer and pay for hepatitis B vaccination
 - b) Provide, maintain, and ensure use of PPE and engineering controls

c) Establish appropriate work practices and decontamination procedures in the oral health care setting to ensure the safety of DHCP (appropriate decontamination, laundry handling, and infectious waste disposal)

d) Establish and provide postexposure medical evaluation and follow-up without any cost to employees

e) Provide appropriate biohazard communication by posting signs, biohazard waste labels, and red containers that indicate infectious waste on-site

(5) Maintain appropriate records

a) Training sessions (participants, trainers, summary of content, evaluations)

b) Employee medical records to include: name and social security number of employee, HBV immunization status, postexposure evaluation from health care professional limited to whether the employee has been informed of the results of the evaluation and has been told about any medical conditions resulting from exposure to blood or other potentially infectious materials that require further evaluation or treatment

c) Employers must keep employee records for the duration of employment plus an additional 30 years

7. Inspections conducted by OSHA

a. Initiated by employee complaints or other interested parties

b. Programmed inspections of randomly selected worksites employing 11 or more people

c. Noncompliance with any provision in the Standard can result in the imposition of fines

Evaluation of Infection Control Programs

Table 10.8 lists program elements with corresponding activities in the evaluation of infection control programs. The websites listed in the Website Information and Resources table provide the oral health care professional with immediate access to current recommendations and regulations concerning infection control.

Table 10.8

Evaluating Infection Control Programs

Program Element	Evaluation Activity
Appropriate immunization of dental health care personnel (DHCP)	Conduct annual review of personnel records to ensure up-to-date immunization.
Assessment of occupational exposures to infectious agents	Report occupational exposures to infectious agents. Document the steps that occurred around the exposure, and plan how such exposure can be prevented in the future.
Comprehensive postexposure management plan and medical follow-up program after occupational exposures to infectious agents	Ensure the postexposure management plan is clear, complete, and available at all times to all DHCP. All staff should understand the plan, which should include toll-free phone numbers for access to additional information.
Adherence to hand hygiene before and after patient care	Observe and document circumstances of appropriate or inappropriate handwashing. Review findings in a staff meeting.
Proper use of personal protective equipment (PPE) to prevent occupational exposures to infectious agents	Observe and document the use of barrier precautions and careful handling of sharps. Review findings in a staff meeting.

Routine and appropriate sterilization of instruments using a biologic monitoring system
Monitor paper log of steam cycle and temperature strip with each sterilization load, and examine results of weekly biologic monitoring. Take appropriate action when failure of sterilization process is noted.

Evaluation and implementation of safer medical devices Conduct an annual review of the exposure control plan, and consider new developments in safer medical devices.

Compliance of water in routine dental procedures with current drinking US Environmental Protection Agency water standards (fewer than 500 colony-forming units [CFU] of heterotrophic water bacteria) Monitor dental water quality as recommended by the equipment manufacturer, using commercial self-contained test kits or commercial water-testing laboratories.

Proper handling and disposal of medical waste Observe the safe disposal of regulated and nonregulated medical waste, and take preventive measures if hazardous situations occur.

Health care–associated infections Assess the unscheduled return of patients after procedures, and evaluate them for an infectious process. A trend might require formal evaluation.

From Centers for Disease Control and Prevention. Guidelines for infection control in dental health-care settings. MMWR. 2003;52(RR-17):1–68.

Legal and Ethical Issues in Disease Prevention

Box 10.3 lists recommendations to address legal and ethical considerations in infection control measures.

Website Information and Resources

Source	Website Address	Description
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Association for Professionals in Infection Control and Epidemiology (APIC)	http://www.apic.org	Infection control organization for health care
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Centers for Disease Control and Prevention (CDC)	http://www.cdc.gov	US government agency that provides resources and recommendations for numerous health and safety topics
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Association for Dental Safety (formerly OSAP)	https://www.myads.org/	Organization that promotes infection control and safety policies to the dental community
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Occupational Safety and Health Administration (OSHA)	http://www.osha.gov	US government agency responsible for ensuring the health and safety of employees at workplaces
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National Institutes for Occupational Safety and Health (NIOSH)	http://www.cdc.gov/niosh	US government agency that provides national and world leadership to prevent illness and injury
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Box 10.3 Legal and Ethical Guidelines in Disease Prevention

Treat all patients regardless of disease status

Practice infection control according to the standard of care

Adhere to the Centers for Disease Control and Prevention and the Occupational Safety and Health Administration guidelines

Adhere to state and federal laws

Use evidence-based protocols
Follow state board of dentistry laws and regulations
Follow expert opinion related to infection control
Stay current of new protocols and guidelines
Abide by US Public Health Service guidelines for employee work restrictions
Adhere to local department of public health guidelines
Participate in contact tracing to help identify people exposed to contagious diseases

Chapter 10 Review Questions

Answers and rationales to review questions are available on this text's accompanying Evolve site. See inside front cover for details.

1. Which of these groups is a regulatory body?
 - a. The Occupational Safety and Health Administration (OSHA)
 - b. The Centers for Disease Control and Prevention (CDC)
 - c. The Association for Dental Safety (ADS)
 - d. The American Dental Association (ADA)
2. Which of these groups provides evidence-based recommendations?
 - a. The Occupational Safety and Health Administration (OSHA)
 - b. The Centers for Disease Control and Prevention (CDC)
 - c. The Environmental Protection Agency (EPA)
 - d. The Food and Drug Administration (FDA)
3. Which of these strains of Hepatitis is NOT bloodborne?
 - a. Hepatitis A
 - b. Hepatitis B
 - c. Hepatitis C
 - d. Hepatitis D
4. Which of these is considered a critical item according to the Spaulding classification?
 - a. Blood pressure cuff
 - b. Mouth mirror
 - c. Radiograph sensor
 - d. Dental hygiene curette
5. Which of these is considered a semicritical item according to the Spaulding classification?
 - a. Blood pressure cuff
 - b. Mouth mirror

- c. Extraction forceps
- d. Dental hygiene curette

6. Which of these is considered a noncritical item according to the Spaulding classification?

- a. Blood pressure cuff
- b. Mouth mirror
- c. Radiograph sensor
- d. Dental hygiene curette

7. A sneeze or a cough is an example of:

- a. Source of microbe or pathogen
- b. A portal of entry
- c. A portal of exit
- d. A vulnerable host

8. All of these are work restrictions for a dental health care worker EXCEPT one. Which is the EXCEPTION?

- a. Conjunctivitis
- b. Pediculosis
- c. Measles
- d. Headache

9. Which best describes engineering controls?

- a. Devices to reduce the risk of exposure incidents
- b. Actions or behaviors that reduce the risk of exposure incidents
- c. Personal protective equipment that reduce the risk of exposure incidents
- d. All of the above

10. All of the following are examples of work practice controls EXCEPT one. Which is the EXCEPTION?

- a. Self-sheathing needles
- b. One-handed scoop technique for recapping a needle
- c. Maintaining a 14–18-inch working distance to reduce droplets
- d. Fulcruming one to four teeth away from tooth being treated

11. According to transmission-based precautions, where would a patient with active tuberculosis be treated?

- a. In the doctor's treatment room
- b. In the dental hygienist's operator
- c. In an airborne isolation room
- d. In a hospital dental clinic

12. According to standard precautions, all of the following are considered potentially infectious EXCEPT one. Which is the EXCEPTION?

- a. Blood
- b. Nonintact skin
- c. Sweat
- d. Mucous membranes

13. Which of the following immunizations are recommended for dental health care workers?

- a. Shingles
- b. Hepatitis A and B
- c. Pneumonia
- d. Polio

14. When should antibody follow-up testing occur for the Hepatitis B vaccine?

- a. 1–2 months after the final injection
- b. 3–4 months after the final injection
- c. 4–6 months after the final injection
- d. 1–2 years after the final injection

15. Hand hygiene should be performed at all of these times EXCEPT one. Which is the EXCEPTION?

- a. Before gloving (donning)
- b. After gloves are removed (doffing)
- c. After using the restroom
- d. Before using the restroom

16. Choose the correct sequence for donning personal protective equipment (PPE).

- a. Protective clothing (lab coat or gown), mask & eyewear, hand hygiene, then gloves
- b. Mask & eyewear, hand hygiene and gloves, then protective clothing (lab coat or gown)
- c. Protective clothing (lab or gown), hand hygiene and gloves, then mask & eyewear
- d. Mask & eyewear, protective clothing (lab coat or gown), then hand hygiene and gloves

17. Which types of gloves should be used during instrument reprocessing?

- a. Nonsterile examination gloves
- b. Sterile surgical gloves
- c. Utility gloves
- d. Overgloves (food handler's gloves)

18. Latex allergy is which type of reaction?

- a. Type I immediate
- b. Type II Cytotoxic
- c. Type III Immune Complex
- d. Type IV delayed

19. Which type of surface disinfectant should be used to wipe clinical contact surfaces that might be contaminated with blood or other potentially infectious material (OPIM)?

- a. High level
- b. Intermediate level
- c. Low level
- d. None of the above

20. Intermediate-level disinfectants carry all of these kill claims EXCEPT one. Which is the EXCEPTION?

- a. C. difficile
- b. HIV
- c. Hepatitis B
- d. S. aureus

21. Which of the following is NOT considered a clinical contact surface?

- a. Chair controls
- b. Bracket tray
- c. Floors
- d. Saliva ejector control

22. Which organization regulates chemical disinfectants?

- a. FDA
- b. CDC
- c. OSHA
- d. EPA

23. Dental Unit Water Lines measured in colony-forming units per milliliter (CFU/mL) must not exceed which level of bacterial contamination?

- a. 50 CFU/mL
- b. 100 CFU/mL
- c. 500 CFU/mL
- d. 1000 CFU/mL

24. What is the first action that should be taken when a percutaneous (sharps injury) happens?

- a. Wash the area with soap and water
- b. Yell for help
- c. Report the injury
- d. Get an HIV test

25. Where should blood-soaked or blood-saturated items should be discarded?

- a. In the trash can at the front desk
- b. In the operatory
- c. In a biohazard container
- d. In the sterilization area trash can

26. How soon should a dental health care worker seek medical care after an exposure incident?

- a. Within one hour
- b. Within two hours
- c. Within four hours
- d. Within 24 hours

27. All of the following are acceptable methods of cleaning for removing debris or bioburden from instruments prior to sterilization EXCEPT one. Which is the EXCEPTION?

- a. Hand scrubbing with a short-handled brush
- b. An ultrasonic bath
- c. An instrument washer
- d. An instrument washer/disinfector

28. Autoclave tape is an example of which type of chemical indicator?

- a. Class I
- b. Class II
- c. Class III
- d. Class VI

29. The best place to store instruments after sterilization is_____?

- a. unwrapped in the sterilization drawer.
- b. unwrapped in the dental operator.
- c. wrapped in a drawer in the operator.
- d. wrapped on a shelf above the autoclave.

30. How frequently should biological monitoring with chemical indicators occur?

- a. At least every hour
- b. At least daily
- c. At least weekly
- d. At least monthly

31. Which type of chemical indicator should be placed inside of a wrapped instrument cassette?

- a. Class I
- b. Class II
- c. Class IV
- d. Class V

32. All of the following methods inhibit the formation of biofilm in dental unit water lines EXCEPT one. Which is the EXCEPTION?

- a. Flushing water lines for 2 minutes
- b. Independent water reservoir
- c. Chemical disinfection of water lines
- d. Sterile-water delivery systems

33. Which governmental agency is responsible for approving the safety and effectiveness of small office sterilizers and ultrasonic units?

- a. EPA
- b. FDA
- c. CDC
- d. OSHA

34. What is the ultimate goal of disease prevention in the oral health care (OHC) setting?

- a. Prevent pathogens from entering the OHC setting.
- b. Sterilize clinical contact surfaces between patients.
- c. Prevent OHC related infections and disease.
- d. Eliminate all microbes in the OHC setting.

35. A dental health care worker has occupationally acquired tuberculosis. Which of the following modes of transmission would be implicated in this case?

- a. Direct contact
- b. Indirect contact
- c. Droplet infection
- d. Airborne infection

36. Which term is used for the practice of treating a patient's blood, body fluids, nonintact skin, and mucous membranes as potentially infectious?

- a. Standard precautions
- b. Pervasive precautions
- c. Universal precautions
- d. Protective precautions

37. Today, effective vaccines are available to prevent which of the following infections?

- a. HIV
- b. Hepatitis C
- c. Hepatitis B
- d. Herpes simplex

38. Which of the following is an example of an engineering control?

- a. Eyewear
- b. Sharps container
- c. Using a one-handed scoop technique for recapping a needle
- d. Handwashing

39. Which of the following is not a link in the chain of infection?

- a. A pathogen in sufficient number
- b. A mode of transmission for the pathogen
- c. An immune host

- d. A portal of entry into a new host
- e. A reservoir for the pathogen

40. A person who is immune/recovered from a disease but can still transmit the disease is called:

- a. A susceptible host
- b. A source patient
- c. A carrier
- d. An at-risk dental health care worker

41. All of the following tasks should be conducted to maintain high-quality dental unit water EXCEPT one. Which is the EXCEPTION?

- a. Use heated water for patient comfort.
- b. Flush water lines for 20–30 seconds between patients.
- c. Use low-level chemical germicides in dental unit water lines.
- d. Periodically test the CFU/mL level in dental unit water.

42. The Centers for Disease Control and Prevention recommends that all dental offices have a designated person who handles infection control training records, monitors infection control policies and procedures, and who communicates updates about infection control policies and procedures. This person is known as the _____.

- a. OSHA monitor
- b. Infection control coordinator
- c. Infection Manager
- d. CDC Monitor

Case A

Jean, a registered dental hygienist (RDH) working through a temporary agency, is assigned a shift at Bright Smiles dental office. When she arrived, Jean was shown her operatory, schedule, and instruments. She was given a brief tour of the office to show her the radiology room, the sterilization lab, and the break room. No one explained any standard operating procedures (SOPs) or showed her where the OSHA manual was located. As Jean was unloading the autoclave, she noticed the instrument pouches were still wet because another staff member opened the door early because she needed extraction instruments right away.

Use case A to answer questions 43 to 46.

43. What should Jean do next?

- a. Go ahead and use the wet instruments.
- b. Wait for the instruments to dry and then use the instruments.
- c. Close the autoclave and restart the cycle.
- d. Rebag the instruments and run a new cycle including a drying cycle.

44. Wet instruments can increase microbial contamination through a process called _____.

- a. sterilization
- b. contamination
- c. indirect contact
- d. wicking

45. How could this practice improve sterilization processes when temporary employees are working?

- a. Have an office manager
- b. Have written Standard Operating Procedures (SOP's)
- c. Have an OSHA manual
- d. Have OSHA training annually

46. Using wet instruments could result in _____?

- a. bacterial contamination
- b. sterility
- c. normal practice
- d. Legionnaire's disease

Case B

Pedro, a registered dental hygienist (RDH), was recently hired at Oceanside Dental Group. On his first day, Pedro was being trained by the lead dental hygienist and Infection Control Coordinator, Sarah. Sarah made sure to explain the schedule, reviewed the OSHA manual, reviewed the walkout procedures, and reviewed all of the infection control policies and procedures located in the Standard Operating Procedures (SOP) manual. As Pedro was turning his treatment room over after dismissing his first patient, he forgot a few things Sarah said regarding the contact time for the disinfectant spray and where the water line treatment tablets were located. He asked another staff member, Cindy, about these questions. Cindy replied, "Sarah goes overboard with infection control. We don't listen to her. Our schedules are so full that we don't have time to do all that stuff." What should Pedro do?

Use case B to answer questions 47 to 50.

47. How long should Pedro wait to seat his next patient after cleaning and disinfecting his operatory?

- a. He should wait for 5 minutes.
- b. Check the product's contact time on the container.
- c. He should not wait; he should seat his next patient immediately to stay on schedule.
- d. He should go ask Sarah.

48. Where could Pedro go to find the location of the water line treatment tablets?

- a. Go ask Sarah.
- b. Go ask the front desk receptionist.
- c. Look in the OSHA manual.
- d. Look in the SOP manual.

49. All of the following are being done correctly with regard to infection control and safety EXCEPT one. Which is the exception?

- a. The office has an infection control coordinator (ICC), Sarah.
- b. The office has written standard operating procedures (SOPs).
- c. The team utilizes a checklist to ensure policies and procedures are being conducted properly.
- d. The office has training for new employees.

50. Where could Pedro find the Safety Data Sheet (SDS) for the chemical disinfectant?

- a. The SOP documents
- b. Go ask Sarah
- c. The OSHA manual
- d. Google

SUGGESTED READINGS

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